

**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan Trunk OSPF MultiVendor

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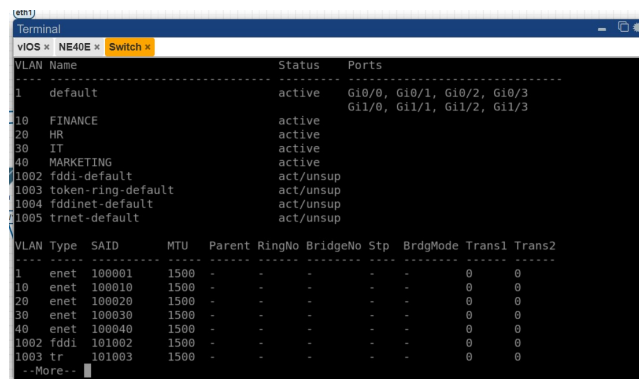
2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1 Konfigurasi Cisco Switch

Langkah Percobaan

1. Masuk ke terminal Cisco Switch.
2. Membuat VLAN.
3. Pastikan VLAN sudah dibuat dan statusnya aktif.

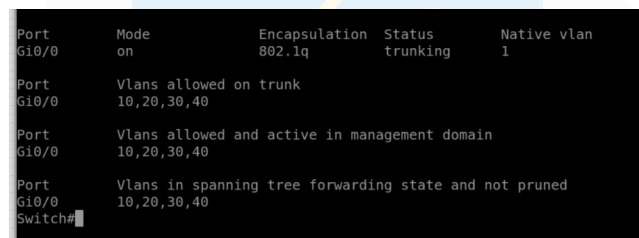


VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	FINANCE	active	
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
40	enet	100040	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0

Gambar 1: Hasil verifikasi VLAN menggunakan perintah `show vlan brief`.

4. Konfigurasi Access Port untuk setiap VLAN.
5. Konfigurasi Trunk ke Cisco Router.



Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gi0/0	10,20,30,40

Port	Vlans allowed and active in management domain
Gi0/0	10,20,30,40

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	10,20,30,40

Gambar 2: Konfigurasi trunk antara Cisco Switch dan Cisco Router.

1.2 Praktikum 2 Konfigurasi Cisco Router

Langkah Percobaan

1. Masuk ke Cisco Router.
2. Konfigurasi Interface Trunk ke Switch.
3. Konfigurasi Subinterface VLAN 10.
4. Konfigurasi Subinterface VLAN 20.
5. Konfigurasi Subinterface VLAN 30.
6. Konfigurasi Subinterface VLAN 40.

7. Konfigurasi DHCP Server untuk VLAN 10.
8. Konfigurasi DHCP Server untuk VLAN 20.
9. Konfigurasi IP Static pada VPCS untuk VLAN 30 dan VLAN 40.
10. Verifikasi Cisco Router.

```

Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#exit
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#exit
Router#show ip
*Jun  6 05:49:00.798: %SYS-5-CONFIG_I: Configured from console by consolein
% Ambiguous command: "show ip in"
Router#show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  up                    up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up                    up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3       unassigned      YES unset  administratively down  down
Router#

```

Gambar 3: Hasil verifikasi interface router menggunakan perintah `show ip interface brief`.

```

GigabitEthernet0/2.20    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3       unassigned      YES unset  administratively down  down
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)        : 0 / 0
  Total addresses                  : 254
  Leased addresses                 : 1
  Pending event                   : none
  1 subnet is currently in the pool :
    Current index  IP address range                Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)        : 0 / 0
  Total addresses                  : 254
  Leased addresses                 : 1
  Pending event                   : none
  1 subnet is currently in the pool :
    Current index  IP address range                Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#

```

Gambar 4: Hasil verifikasi DHCP pool menggunakan perintah `show ip dhcp pool`.

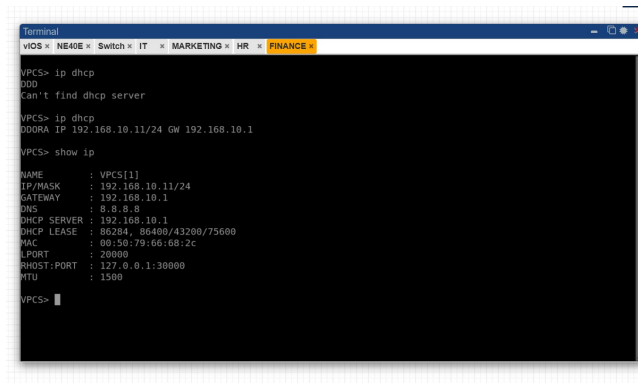
11. Pengujian dari VPCS.

```

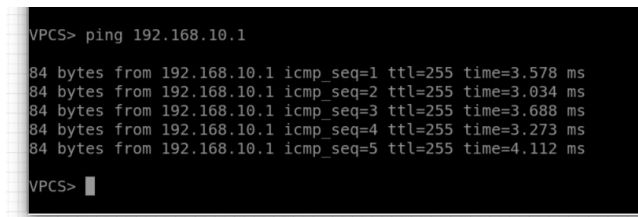
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.2c Jun 07 2026 05:48 AM    Automatic
192.168.20.11    0100.5079.6668.2d Jun 07 2026 05:46 AM    Automatic
Router#

```

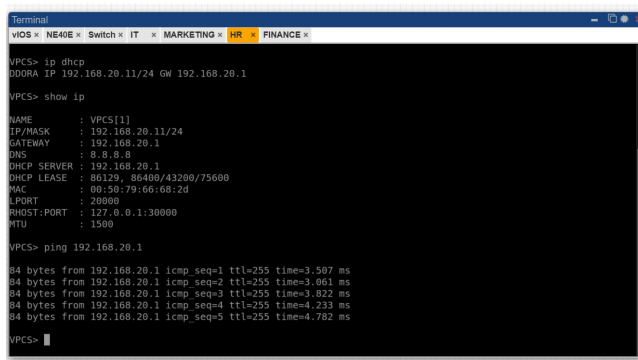
Gambar 5: Hasil verifikasi DHCP binding menggunakan perintah `show ip dhcp binding`.



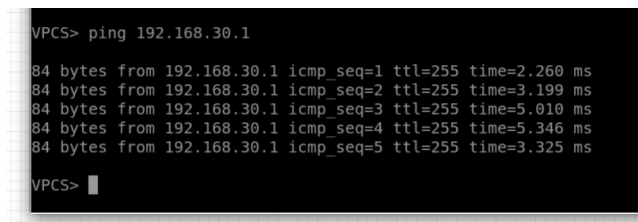
Gambar 6: Pengujian DHCP dan verifikasi alamat IP pada VLAN 10 (Finance).



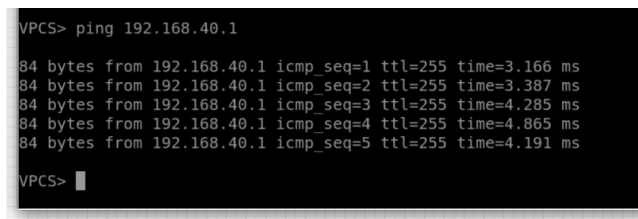
Gambar 7: Pengujian konektivitas ke gateway 192.168.10.1 dari VLAN 10 (Finance).



Gambar 8: Pengujian DHCP, verifikasi IP, dan konektivitas gateway pada VLAN 20 (HR).



Gambar 9: Pengujian konektivitas ke gateway 192.168.30.1 dari VLAN 30 (IT).



Gambar 10: Pengujian konektivitas ke gateway 192.168.40.1 dari VLAN 40 (Marketing).

```

VPCS> ping 192.168.20.11
Pinging 192.168.20.11: icmp_seq=1 ttl=63 time=11.444 ms
Pinging 192.168.20.11: icmp_seq=2 ttl=63 time=0.351 ms
Pinging 192.168.20.11: icmp_seq=3 ttl=63 time=0.029 ms
Pinging 192.168.20.11: icmp_seq=4 ttl=63 time=0.128 ms
Pinging 192.168.20.11: icmp_seq=5 ttl=63 time=5.692 ms

VPCS> ping 192.168.30.10
Pinging 192.168.30.10: icmp_seq=1 ttl=63 time=10.884 ms
Pinging 192.168.30.10: icmp_seq=2 ttl=63 time=4.367 ms
Pinging 192.168.30.10: icmp_seq=3 ttl=63 time=0.392 ms
Pinging 192.168.30.10: icmp_seq=4 ttl=63 time=4.252 ms
Pinging 192.168.30.10: icmp_seq=5 ttl=63 time=0.587 ms

VPCS> ping 192.168.40.10
Bad command: "192.168.40.10". Use ? for help.

VPCS> ping 192.168.40.10
Pinging 192.168.40.10: icmp_seq=1 ttl=63 time=10.180 ms
Pinging 192.168.40.10: icmp_seq=2 ttl=63 time=4.165 ms
Pinging 192.168.40.10: icmp_seq=3 ttl=63 time=5.664 ms
Pinging 192.168.40.10: icmp_seq=4 ttl=63 time=4.628 ms
Pinging 192.168.40.10: icmp_seq=5 ttl=63 time=4.733 ms

VPCS>

```

Gambar 11: Pengujian komunikasi antar VLAN dari VLAN 10.

```

Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.2c Jun 07 2026 05:48 AM Automatic
192.168.20.11    0100.5079.6668.2c Jun 07 2026 05:46 AM Automatic
Router#

```

Gambar 12: Verifikasi ulang DHCP binding setelah seluruh pengujian selesai dilakukan.

1.3 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

Langkah Percobaan

1. Konfigurasi LAN Huawei

```

VPCS>
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
monitor[2952]: [SVRPTIME(2952) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
location[2957]: [SVRPTIME(2957) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
location[2959]: [SVRPTIME(2959) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
boot[2958]: [SVRPTIME(2958) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
start fail first time.iret=0
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:09 UTC):
location[3861]: [SVRPTIME(3861) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:09 UTC):
location[3883]: [SVRPTIME(3883) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date

```

Gambar 13: Proses inisialisasi dan aktivasi router Huawei.

2. Konfigurasi IP VPC LAN Huawei

```

Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.50.10/24
GATEWAY   : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:4b
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS>

```

Gambar 14: Verifikasi alamat IP pada VPC LAN Huawei.

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=15.015 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.436 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.871 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.959 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.008 ms

VPCS>
```

Gambar 15: Pengujian konektivitas ke gateway Huawei 192.168.50.1.

3. Konfigurasi IP Link Cisco <-> Huawei Link 1
4. Konfigurasi IP Link Cisco <-> Huawei Link 2
5. Konfigurasi IP Link Huawei <-> MikroTik
6. Konfigurasi IP Link Cisco <-> MikroTik
7. Verifikasi Interface

```
% Ambiguous command: "show ip i"
Router#show ip interface brief

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.11.1      YES manual up          up
GigabitEthernet0/1  10.10.30.1      YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1   YES manual up          up
GigabitEthernet0/2.20  192.168.20.1   YES manual up          up
GigabitEthernet0/2.30  192.168.30.1   YES manual up          up
GigabitEthernet0/2.40  192.168.40.1   YES manual up          up
GigabitEthernet0/3    10.10.10.1      YES manual up          up
Router#
```

Gambar 16: Verifikasi status interface pada router Cisco menggunakan perintah show ip interface brief.

```
[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(L): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0   10.10.10.2/30    up        up        --
Ethernet1/0/0.4094  128.1.138.137/16 up        up        l3vpn
Ethernet1/0/1    unassigned       up        down      --
Ethernet1/0/1.4094  128.1.138.137/16 up        up        l3vpn
Ethernet1/0/2    192.168.50.1/24  up        up        --
Ethernet1/0/2.4094  128.1.138.137/16 up        up        l3vpn
Ethernet1/0/3    10.10.20.1/30    up        up        --
Ethernet1/0/3.4094  128.1.138.137/16 up        up        l3vpn
Ethernet1/0/4    10.10.11.2/30    up        up        --
Ethernet1/0/4.4094  128.1.138.137/16 up        up        l3vpn
.... More ....
```

Gambar 17: Verifikasi status interface pada router Huawei menggunakan perintah display ip interface brief.

```
[admin@MikroTik] /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >
```

Gambar 18: Verifikasi status interface pada router MikroTik.

8. Verifikasi Ping Antar Router

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

Gambar 19: Pengujian konektivitas antarrouter dari router Cisco.

```

Terminal
vIOS x NE40E x Switch x MARKETING x IT x HR x FINANCE x VPC x Mikrotik x

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 3/3/3 ms

[~HUAWEI]

```

Gambar 20: Pengujian konektivitas antarrouter dari router Huawei.

```

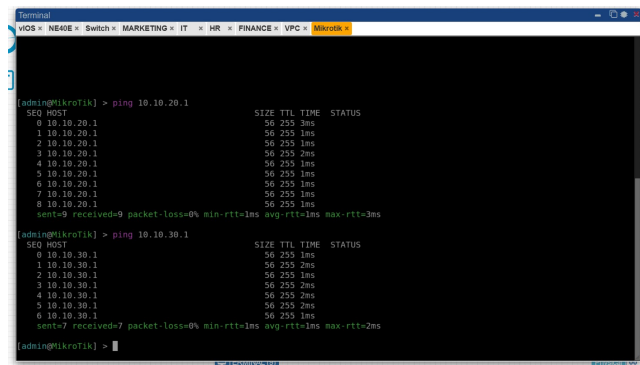
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=3 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/3 ms

[~HUAWEI]

```

Gambar 21: Pengujian konektivitas lanjutan dari router Huawei.



Gambar 22: Pengujian konektivitas antarrouter dari router MikroTik.

1.4 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

Langkah Percobaan

1. Konsep Routing OSPF pada Topologi

Link	Cost	Fungsi
Cisco ↔ Huawei Link 1	10	Jalur utama
Cisco ↔ Huawei Link 2	10	Jalur utama
Cisco ↔ MikroTik	100	Jalur backup
Huawei ↔ MikroTik	100	Jalur backup

Gambar 23: Topologi jaringan yang digunakan pada implementasi OSPF multivendor.

2. Network yang Diiklankan ke OSPF

Cisco Router

Network	Keterangan
10.10.10.0/30	Link Cisco-Huawei 1
10.10.11.0/30	Link Cisco-Huawei 2
10.10.30.0/30	Link Cisco-MikroTik
192.168.10.0/24	VLAN 10 Finance
192.168.20.0/24	VLAN 20 HR
192.168.30.0/24	VLAN 30 IT
192.168.40.0/24	VLAN 40 Marketing

Gambar 24: Daftar network yang dikonfigurasi untuk diiklankan melalui OSPF.

Huawei Router

Network	Keterangan
10.10.10.0/30	Link Huawei-Cisco 1
10.10.11.0/30	Link Huawei-Cisco 2
10.10.20.0/30	Link Huawei-MikroTik
192.168.50.0/24	LAN Huawei

Mikrotik

Network	Keterangan
10.10.20.0/30	Link MikroTik-Huawei
10.10.30.0/30	Link MikroTik-Cisco

Gambar 25: Verifikasi network yang telah berhasil diiklankan ke OSPF.

3. Konfigurasi OSPF pada Cisco Router

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#

```

Gambar 26: Verifikasi konfigurasi OSPF pada router Cisco menggunakan perintah `show ip protocols`.

4. Konfigurasi OSPF pada Huawei Router

```

[~HUAWEI] Ethernet1/0/3 quit
[~HUAWEI] commit
[~HUAWEI] display ospf peer

(M) Indicates MADJ neighbor

    OSPF Process 1 with Router ID 2.2.2.2
      Neighbors
    Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
    Router ID: 1.1.1.1          Address: 10.10.10.1
    State: Full                Mode: Nbr is Slave      Priority: 1
    DR: 10.10.10.1             BDR: 10.10.10.2         MTU: 1500
    Dead timer due in 38 sec
    Retrans timer interval: 5
    Neighbor is up for 00h01m59s
    Neighbor Up Time : 2026-06-06 06:29:29
    Authentication Sequence: [ 0 ]

    Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
    Router ID: 1.1.1.1          Address: 10.10.11.1
    State: Full                Mode: Nbr is Slave      Priority: 1
    DR: 10.10.11.1             BDR: 10.10.11.2         MTU: 1500
    Dead timer due in 37 sec
    Retrans timer interval: 5
    Neighbor is up for 00h01m59s
    Neighbor Up Time : 2026-06-06 06:29:29
    Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 27: Verifikasi OSPF neighbor pada router Huawei menggunakan perintah `display ospf peer`.

```

Terminal
vOS > NEGE > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
.....
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.30.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.50.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.50.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/0
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]

```

Gambar 28: Status hubungan neighbor OSPF antara router Huawei dan Cisco pada dua link utama.

```

Terminal
vOS > NEGE > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >
Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 120.1.130.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
120.1.130.137/32 0 Direct 120.1.130.137 120.1.130.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]

```

Gambar 29: Tabel routing OSPF pada router Huawei menggunakan perintah `display ospf routing`.

5. Konfigurasi OSPF pada MikroTik

```

[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="ExStart" state-changes=3 ls-retransmits=0 ls-requests=0 db-summaries=0
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=5s
[admin@Mikrotik] >

```

Gambar 30: Konfigurasi OSPF pada router MikroTik.

```

[admin@Mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
0 - blackhole, 0 - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@Mikrotik] >

```

Gambar 31: Verifikasi routing table OSPF pada router MikroTik.

6. Verifikasi OSPF Neighbor

```

Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:36 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:38 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:34 10.10.10.2 GigabitEthernet0/3
Router#

```

Gambar 32: Verifikasi OSPF neighbor pada router Cisco.

```

vlos > display ospf peer
Switch = MARKETING > IT > HR > FINANCE > VPC > Mikrotik >
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/3)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00h05m27s
Neighbor Up Time : 2026-06-06 06:29:29
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/2)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h05m11s
Neighbor Up Time : 2026-06-06 06:33:45
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00h05m27s
Neighbor Up Time : 2026-06-06 06:29:29
Authentication Sequence: [ 0 ]
[~HUAWEI]

```

Gambar 33: Verifikasi OSPF neighbor pada router Huawei.

```

[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5 is-retransmits=0 is-requests=0 db-summaries=0
  adjacency=lm485
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 is-retransmits=0 is-requests=0 db-summaries=0
  adjacency=lm485
[admin@Mikrotik] >

```

Gambar 34: Verifikasi OSPF neighbor pada router Mikrotik.

7. Verifikasi Routing Table

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:18 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#

```

Gambar 35: Verifikasi routing table pada router Cisco menuju jaringan LAN Huawei.

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O 10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:55, GigabitEthernet0/0
  [110/110] via 10.10.10.2, 00:04:55, GigabitEthernet0/3
O 192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:40, GigabitEthernet0/0
  [110/11] via 10.10.10.2, 00:06:40, GigabitEthernet0/3
Router#

```

Gambar 36: Verifikasi routing table hasil pembelajaran rute OSPF.

```

vlos > display ip routing-table
Switch = MARKETING > IT > HR > FINANCE > VPC > Mikrotik >
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.20.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.50.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.50.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]

```

Gambar 37: Verifikasi routing table pada router Huawei.

```

admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
      - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 Ado 10.10.10.0/30    10.10.20.1 110
1 Ado 10.10.11.0/30    10.10.20.1 110
2 Ado 192.168.10.0/24   10.10.30.1 110
3 Ado 192.168.20.0/24   10.10.30.1 110
4 Ado 192.168.30.0/24   10.10.30.1 110
5 Ado 192.168.40.0/24   10.10.30.1 110
6 Ado 192.168.50.0/24   10.10.20.1 110
admin@MikroTik] >

```

Gambar 38: Verifikasi routing table OSPF pada router MikroTik

8. Verifikasi Koneksi End-to-End

```

Terminal
vIOS < NE40E < Switch < MARKETING < IT < HR < FINANCE < VPC < Mikrotik <
VPCS> ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=12.509 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=4.206 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=10.011 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=5.909 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=6.551 ms
VPCS>

```

Gambar 39: Pengujian konektivitas ke gateway Huawei dari VLAN 10 (Finance).

```

Terminal
vIOS < NE40E < Switch < MARKETING < IT < HR < FINANCE < VPC < Mikrotik <
VPCS> ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=0.120 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=9.276 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=9.645 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=7.110 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=9.282 ms
VPCS>

```

Gambar 40: Pengujian konektivitas ke gateway Huawei dari VLAN 20 (HR).

```

Terminal
vIOS < NE40E < Switch < MARKETING < IT < HR < FINANCE < VPC < Mikrotik <
VPCS> ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=7.291 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=10.006 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=4.095 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=6.309 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=5.930 ms
VPCS>

```

Gambar 41: Pengujian konektivitas ke gateway Huawei dari VLAN 30 (IT).

```

Terminal
vIOS < NE40E < Switch < MARKETING < IT < HR < FINANCE < VPC < Mikrotik <
VPCS> ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=7.259 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=7.798 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=8.738 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=5.976 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=6.645 ms
VPCS>

```

Gambar 42: Pengujian konektivitas ke gateway Huawei dari VLAN 40 (Marketing).

```

VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=5.763 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=7.343 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.581 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=7.012 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=7.432 ms
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=6.548 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=6.026 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=7.515 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.380 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=6.237 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.878 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=6.539 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.228 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=9.168 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=8.209 ms

```

Gambar 43: Pengujian komunikasi dari PC LAN Huawei menuju jaringan VLAN Cisco (bagian 1).

```

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=7.791 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.405 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=6.698 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=9.075 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=8.409 ms
VPCS>

```

Gambar 44: Pengujian komunikasi dari PC LAN Huawei menuju jaringan VLAN Cisco (bagian 2).

1.5 Praktikum 5 Pengujian Failover OSPF

Langkah Percobaan

1. Kondisi normal: Dua link Cisco-Huawei aktif

```

VPCS> trace 192.168.50.10
traceroute to 192.168.50.10, 8 hops max
 1 192.168.50.10    0.001 ms
VPCS>

```

Gambar 45: Pengujian konektivitas pada kondisi normal saat kedua link Cisco-Huawei aktif menggunakan *trace* dari PC LAN Huawei.

2. Kondisi 1 link Cisco-Huawei mati

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1    4.828 ms  6.104 ms  4.002 ms
 2 10.10.11.2      6.361 ms  5.135 ms  3.211 ms
 3 *192.168.50.10 11.337 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 46: Pengujian jalur komunikasi setelah salah satu link Cisco-Huawei dinonaktifkan.

3. Kondisi dua link Cisco-Huawei mati

```

VPCS> NEAR< Switch> MARKETING< IT< HR< FINANCE< VPC< Mikrotik<
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#
Jan 6 06:52:12.869: NDSPP-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Inter-
face down or detached
% Ambiguous command: "e"
Router(config)#
Jan 6 06:52:14.038: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
Jan 6 06:52:15.038: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#exit
Router#
Jan 6 06:52:17.331: %SYS-5-CONFIG-I: Configured from console by consoleconf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual administratively down down
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual administratively down down
Router(config)#

```

Gambar 47: Kondisi jaringan ketika kedua link Cisco-Huawei dinonaktifkan (*shutdown*).

4. Mengaktifkan kembali dua link Cisco-Huawei

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.967 ms  4.179 ms  4.396 ms
 2  10.10.11.2     7.091 ms  5.054 ms  6.015 ms
 3  *192.168.50.10  8.867 ms  (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 48: Pengujian konektivitas setelah kedua link Cisco-Huawei diaktifkan kembali.

2 Analisis Hasil Percobaan

2.1 Konfigurasi Cisco Switch

Pada praktikum ini berhasil dibuat empat VLAN pada Cisco Switch, yaitu VLAN 10 (FINANCE), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (MARKETING). Berdasarkan hasil verifikasi menggunakan perintah `show vlan brief`, seluruh VLAN terdeteksi dalam keadaan *active*. Setelah itu, setiap port switch dikonfigurasi sebagai *access port* dan ditempatkan pada VLAN yang sesuai. Selanjutnya, antarmuka GigabitEthernet0/0 dikonfigurasi sebagai *trunk port* menggunakan standar IEEE 802.1Q untuk menghubungkan switch dengan router. Hasil pengecekan menunjukkan bahwa seluruh VLAN dapat melewati trunk tersebut dengan baik, sehingga proses segmentasi jaringan telah berjalan sesuai desain yang direncanakan.

2.2 Konfigurasi Cisco Router

Konfigurasi router dilakukan menggunakan metode *Router-on-a-Stick*, yaitu dengan membuat beberapa subinterface pada satu antarmuka fisik untuk melayani komunikasi antar VLAN. Masing-masing VLAN diberikan alamat gateway 192.168.10.1/24, 192.168.20.1/24, 192.168.30.1/24, dan 192.168.40.1/24. Hasil verifikasi melalui perintah `show ip interface brief` menunjukkan seluruh subinterface berada pada status *up/up*, yang menandakan konfigurasi berhasil diterapkan. Selain itu, layanan DHCP yang dikonfigurasi pada VLAN 10 dan VLAN 20 berhasil mendistribusikan alamat IP secara otomatis kepada host yang terhubung. Hal ini dibuktikan melalui hasil `show ip dhcp binding` yang menampilkan alamat IP yang telah diberikan kepada klien. Pengujian konektivitas dari setiap VPCS menuju gateway masing-masing juga berhasil dilakukan tanpa kendala. Uji komunikasi antar VLAN menunjukkan bahwa proses *inter-VLAN routing* telah berjalan dengan baik sehingga perangkat dari jaringan yang berbeda dapat saling berkomunikasi.

2.3 Konfigurasi Huawei Router dan IP Address Antarrouter

Pada praktikum ini dilakukan pengalamatan IP pada beberapa link antarrouter yang menghubungkan perangkat Cisco, Huawei, dan MikroTik. Koneksi antarrouter dibangun menggunakan jaringan point-to-point dengan subnet /30 untuk menghemat penggunaan alamat IP. Selain itu, router Huawei juga dikonfigurasi untuk melayani jaringan LAN dengan subnet 192.168.50.0/24. Berdasarkan hasil verifikasi antarmuka pada ketiga router, seluruh interface yang digunakan berada dalam kondisi aktif dan dapat beroperasi dengan normal. Pengujian menggunakan perintah *ping* dari masing-masing router menunjukkan bahwa seluruh perangkat dapat saling terhubung tanpa kehilangan paket. Hasil tersebut membuktikan bahwa konfigurasi IP addressing dan konektivitas fisik antarrouter telah berhasil sebelum proses routing dinamis diterapkan.

2.4 Konfigurasi OSPF Multivendor dan OSPF Cost

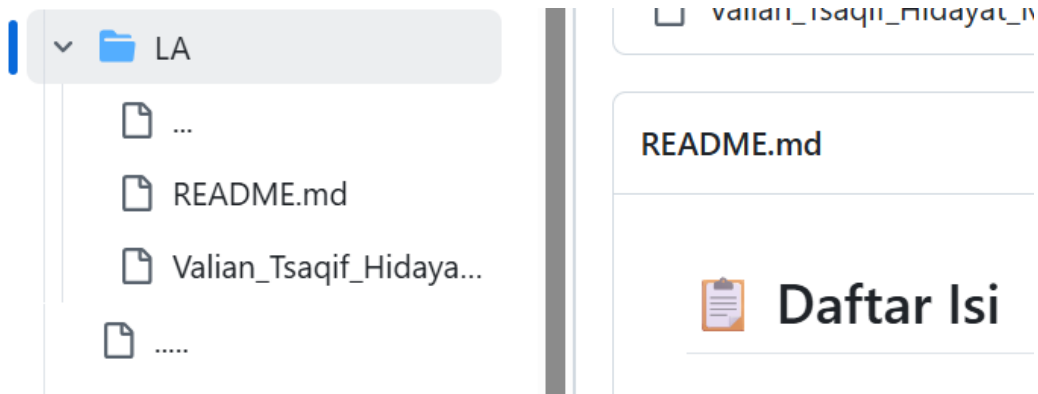
Protokol routing OSPF dikonfigurasi pada router Cisco, Huawei, dan MikroTik dalam area yang sama, yaitu Area 0. Pada topologi ini, link utama antara Cisco dan Huawei diberikan nilai cost yang lebih kecil dibandingkan jalur yang melewati MikroTik sehingga OSPF akan memprioritaskan jalur utama sebagai rute terbaik. Setelah konfigurasi selesai dilakukan, proses pembentukan neighbor OSPF berhasil terbentuk pada seluruh router dengan status yang menunjukkan hubungan adjacency berjalan normal. Tabel routing pada masing-masing perangkat memperlihatkan bahwa seluruh jaringan tujuan telah dipelajari melalui OSPF. Pada router Cisco juga terlihat adanya lebih dari satu jalur menuju jaringan tertentu yang menunjukkan kemampuan OSPF dalam melakukan pemilihan rute terbaik dan *load balancing* apabila tersedia beberapa jalur dengan biaya yang setara. Pengujian koneksi end-to-end dari berbagai VLAN menuju jaringan LAN Huawei serta sebaliknya berhasil dilakukan, yang menandakan proses pertukaran informasi routing telah berjalan dengan baik.

2.5 Pengujian Failover OSPF

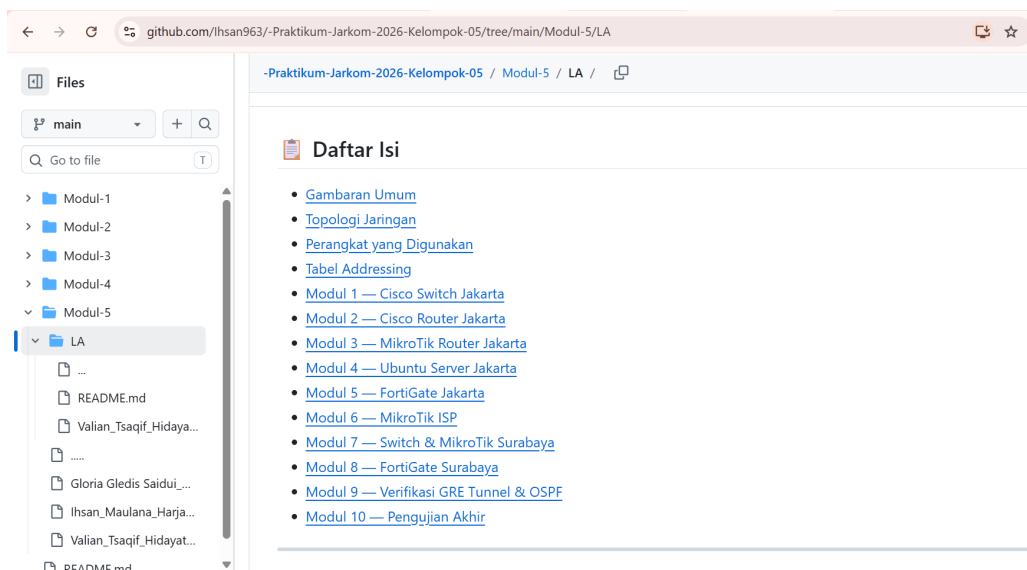
Pengujian failover dilakukan untuk mengamati kemampuan OSPF dalam menyesuaikan jalur ketika terjadi gangguan pada jaringan. Pada kondisi awal, saat kedua link Cisco-Huawei aktif, paket data dikirim melalui jalur utama yang memiliki nilai cost paling rendah sehingga menghasilkan waktu respons yang lebih cepat. Ketika salah satu link utama dimatikan, OSPF secara otomatis melakukan proses konvergensi dan tetap mempertahankan konektivitas dengan menggunakan link yang masih tersedia. Selanjutnya, saat kedua link utama dinonaktifkan, lalu lintas jaringan berhasil dialihkan ke jalur cadangan yang melewati router MikroTik. Meskipun terjadi peningkatan waktu tempuh paket karena melewati jalur dengan cost yang lebih besar, komunikasi antarjaringan tetap dapat berlangsung. Setelah kedua link utama diaktifkan kembali, OSPF kembali melakukan perhitungan rute dan secara otomatis memilih jalur utama tanpa memerlukan konfigurasi tambahan dari administrator. Hasil ini menunjukkan bahwa mekanisme failover dan recovery pada OSPF bekerja sesuai dengan yang diharapkan.

3 Hasil Tugas Modul

Untuk tugas modul, seluruh konfigurasi, dokumentasi, dan pembahasan telah diunggah ke repositori GitHub kelompok dalam bentuk file `README.md`.



Gambar 49: Dokumentasi README.md pada repositori GitHub



Gambar 50: Dokumentasi README.md pada repositori GitHub

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi jaringan multivendor yang melibatkan perangkat Cisco, Huawei, dan MikroTik berhasil diimplementasikan dengan baik. VLAN dan trunking menggunakan standar 802.1Q mampu melakukan segmentasi jaringan sesuai kebutuhan, sedangkan metode *Router-on-a-Stick* berhasil mendukung komunikasi antar VLAN. Selain itu, layanan DHCP pada Cisco Router dapat mendistribusikan alamat IP secara otomatis kepada klien. Implementasi OSPF Area 0 pada ketiga router juga berjalan dengan baik, ditunjukkan oleh terbentuknya hubungan *neighbor* dan pertukaran informasi routing secara normal. Pengujian failover membuktikan bahwa OSPF mampu mengalihkan jalur komunikasi ke rute cadangan saat terjadi gangguan dan kembali menggunakan jalur utama ketika koneksi pulih. Dengan demikian, praktikum ini menunjukkan bahwa penerapan OSPF dengan pengaturan cost yang tepat serta desain jaringan yang redundan dapat meningkatkan keandalan, ketersediaan, dan efisiensi jaringan.

5 Lampiran

5.1 Dokumentasi saat praktikum

